

Customer Shopping Behavior Analysis

1.Project Overview

This project analyzes customer shopping behavior using transactional data from 5050 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2.Dataset Summary

- Rows: 5050
- Columns: 19
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 556 values in Purchase Amount, 370 values in Size, 601 values in Review Rating and 548 in Previous Purchases columns.

3. Exploratory Data Analysis using Python

I began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info() to check structure and .describe() for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	Payment Method	Frequency of Purchases
count	5050.000000	5050.000000	5050	5050	5050	4494.000000	5050	4680	5050	5050	4449.000000	5050	5050	5050	4502.0000	5050	5050
unique	NaN	NaN	3	30	5	NaN	54	4	25	5	NaN	2	6	2	NaN	6	7
top	NaN	NaN	Male	Shirt	Clothing	NaN	New York	M	Black	Spring	NaN	No	Standard	No	NaN	Credit Card	Monthly
freq	NaN	NaN	3018	320	2130	NaN	337	1981	404	1280	NaN	3462	1213	2789	NaN	954	946
mean	2519.570891	44.150495	NaN	NaN	NaN	144.765236	NaN	NaN	NaN	NaN	3.668195	NaN	NaN	NaN	25.2211	NaN	NaN
std	1470.402964	15.282328	NaN	NaN	NaN	275.590101	NaN	NaN	NaN	NaN	0.865357	NaN	NaN	NaN	14.5211	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	10.120000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN	NaN	0.0000	NaN	NaN
25%	1252.250000	31.000000	NaN	NaN	NaN	41.000000	NaN	NaN	NaN	NaN	3.000000	NaN	NaN	NaN	13.0000	NaN	NaN
50%	2499.500000	44.000000	NaN	NaN	NaN	65.000000	NaN	NaN	NaN	NaN	3.700000	NaN	NaN	NaN	25.0000	NaN	NaN
75%	3740.750000	57.000000	NaN	NaN	NaN	89.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	38.0000	NaN	NaN
max	5099.000000	70.000000	NaN	NaN	NaN	1499.760000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	50.0000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the Purchase Amount, Size, Review Rating and Previous Purchases columns using the mean and mode rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Database Integration:** Connected Python script to Microsoft SQL Server and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in Microsoft SQL Server to answer key business questions:

1. Revenue by Gender – Compared total revenue generated by male vs. female customers vs. others.

	gender	revenue
1	Other	168246
2	Male	313392
3	Female	249427

2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
224	4353	145
225	4272	145
226	4441	708
227	4777	145
228	4789	145
229	4246	145
230	4608	145
231	4698	643
232	4813	145
233	4334	837
234	4114	1265
235	4600	1461
236	4864	145
237	4471	145
238	4852	464
239	4939	586
240	4862	145
241	4039	145

Query executed successfully.

3. Top 5 Products by Rating – Found products with the highest average review ratings.

	item_purchased	Average Product Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.81
4	Hat	3.8
5	T-shirt	3.79

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	(No column name)
1	Standard	233.21
2	Express	240.67

5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

	subscription_status	total_customers	avg_spend	total_revenue
	Yes	1588	189.62	301121.95
	No	3462	124.19	429942.49

6. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
1	Headphones	51
2	Laptop	51
3	Phone	51
4	Hat	50
5	Sneakers	49

7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	Number of Customers
1	Returning	818
2	Loyal	4135
3	New	97

8. Top 3 Products per Category – Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	174
2	2	Accessories	Belt	164
3	3	Accessories	Sunglasses	164
4	1	Clothing	Shirt	228
5	2	Clothing	Blouse	172
6	3	Clothing	Pants	171
7	1	Electronics	Phone	64
8	2	Electronics	Watch	63
9	3	Electronics	Bag	53
10	1	Footwear	Sandals	163
11	2	Footwear	Shoes	153
12	3	Footwear	Boots	146
13	1	Outerwear	Jacket	166
14	2	Outerwear	Coat	162

9. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status	repeat_buyers
1	Yes	1466
2	No	3077

10. Revenue by Age Group – Calculated total revenue contribution of each age group.

	age_group	total_revenue
1	Young Adult	192980.3
2	Senior	190443.04
3	Middle-aged	181878.91
4	Adult	165762.19

5. Dashboard in Power BI

Finally, I built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.